

Caesar Cipher Instruction Manuel

The Caesar Cipher is one of the simplest and widely known encryption techniques in the world. This encryption method substitutes each character in the text by shifting it by a specified amount, for example shifting the letter A by 3 would output the letter D. Caesar Cipher is often used as part of a larger system and still has modern day applications.

How the program functions

```
def Caesar_Cipher(text, shift):
    result = ""
    for char in text:
        if char.isalpha():
            shift_start = ord('A') if char.isupper() else ord('a')
            result += chr(shift_start + (ord(char) - shift_start + shift) % 26)
        else:
            result += char
    return result

# Main program starts here

# Gather user inputs
while True:
    Msg = input("Enter a message you would like to encrypt or decrypt: ")
    Shift = int(input("Enter a value for the characters to be shifted (e.g., 3): "))

    Mode = input("Type 'encrypt' to encode or 'decrypt' to decode: ").strip().lower()

    if not Mode.isalpha():
        print()
        print("Please enter 'encrypt' or 'decrypt' only.")
        continue

    # Perform required calculations and display results

    if Mode == "encrypt":
        encrypted_message = Caesar_Cipher(Msg, Shift)
        print("Encrypted message:", encrypted_message)
    elif Mode == "decrypt":
        decrypted_message = Caesar_Cipher(Msg, -Shift)
        print("Decrypted message:", decrypted_message)
    else:
        print("Please enter 'encrypt' or 'decrypt'.")
        continue

    Continue = input("Would you like to encrypt/decrypt another message? (yes/no: ").strip().lower()
    if Continue != "yes":
        print("Thank you for using this encryption program!")
        break
```

- The program will ask you to input a message
- Choose a shift value which determines how many places each letter moves on the alphabet
- Determine if you wish to encrypt or decrypt the message
- The program will then check for upper and lowercase letters to shift separately
- Then performs the shift for each character, attaching any non-characters where they were in the original

Example

Step 1	Open the script or Run the program in a terminal
Step 2	Enter a message you would like to encrypt or decrypt: Hello World!
Step 3	Enter a value for the characters to be shifted (e.g., 3): 5
Step 4	Type 'encrypt' to encode or 'decrypt' to decode: Encrypt
Output	Encrypted message: Mjqqt Btwqi!
Decrypted	Decrypted message: Hello World!